INCH-POUND

MIL-M-28787/248A(SH)
AMENDMENT 2
18 December 1991
SUPERSEDING
AMENDMENT 1
6 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, NON-INVERTING, DIGITAL, KEY CODE AMC

This amendment forms a part of Military Specification MIL-M-28787/248A(EC), dated 21 July 1981, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4 and 5

TABLE I: Delete and substitute new table I as printed on pages 5 and 6 of this amendment

PAGE 6

* 4.4.2, second line. Delete "A, B, or C" and substitute "A and C".

* 4 4.2, third line: Delete "A, B, C" and substitute "A, C".

* 4.5.1, third line Delete "A through V" and subsutute "D through Q, and U"

* 4.5.1, fifth and sixth lines: Delete "C through H, L through Q, and N" and substitute "D through Q, and U".

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* 4.5.3, third and fourth lines: Delete "C through H, L through Q, and U" and substitute "D through Q, and U".

* 4.5.3, sixth line: Delete "C through V" and substitute "D through Q, and U"

6.4, second line: Delete "100°C" and substitute "85°C".

6.4, Maximum quiescent power dissipation: For class I, delete "5 4" and substitute "2.7" and for class II, delete "5.4" and substitute "4 5".

* 6.4, sixth line under Input current: Delete "Low level (V_{IL}=0V, V_{DD}=18V)" and substitute "Low level (V_{IL}=0V, V_{DD}=15V)".

* 6 4, seventh line under Input current: Delete "High level (V_{IH}=18V, V_{DD}=18V)" and substitute "High level (V_{IH}=15V, V_{DD}=18V)".

PAGE 7

* Delete and substitute new page 7 as printed on page 7 of this amendment.

PAGE 9

* FIGURE 2, Item P07: Delete "18.0V ± 1%" and substitute "15.0V ± 1%"

PAGE 10

* FIGURE 3: Delete and substitute new figure 3 as printed on page 8 of this amendment.

PAGE 15

FIGURE 6: Delete and substitute new figure 6 as printed on page 9 of this amendment.

PAGE 18

FIGURE 7: Delete and substitute new figure 7 as printed on page 10 of this amendment.

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PAGE 20

FIGURE 8: Delete and substitute new figure 8 as printed on page 11 of this amendment.

PAGE 23

FIGURE 9: Delete and substitute new figure 9 as printed on page 12 of this amendment.

PAGE 26

* FIGURE 10, Item P07: Delete "18 V ± 1%" and substitute "15.0V ± 1%".

PAGE 28

* FIGURE 11, Item P07: Delete "18 V ± 1%" and substitute "15.0V ± 1%".

PAGE 29

* FIGURE XI, Test condition L1, Max column Delete "12.0" and substitute "822.0".

PAGES 31 AND 32

* TABLE XII, Test conditions M1 through M18, Min column: Delete "20" (two places) and substitute "6" (two places).

PAGES 32 AND 33

* TABLE XII, Test conditions M19 through M36, Min column: Delete "40" (four places) and substitute "12" (four places).

PAGES 34 AND 35

* TABLE XIII, Test conditions N1 through N18, Min column: Delete "8" (two places) and substitute "2" (two places).

PAGES 35 AND 36

* TABLE XIII, Test conditions N19 through N36, Min column: Delete "14" (four places) and substitute "4" (four places).

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AMENDMENT 2

PAGES 37 AND 38

* TABLE XIV, Test conditions P1 through P18, Min column: Delete "16" (two places) and substitute "5" (two places).

PAGES 38 AND 39

* TABLE XIV, Test conditions P19 through P36, Min column. Delete "21" (four places) and substitute "6" (four places).

PAGES 40 AND 41

* TABLE XV, Test conditions Q1 through Q18, Min column: Delete "11" (two places) and substitute "3" (two places).

PAGES 41 AND 42

* TABLE XV, Test conditions Q19 through Q36, Min column: Delete "13" (four places) and substitute "4" (four places).

PAGE 49

* TABLE XVII, Test condition S14, C to pin column, second line: Delete "36".

* TABLE XVII, Test condition S17, C to pin column, second line: Delete "19".

* TABLE XVII, Test condition S18, C to pin column, second line: Delete "18".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH

(Project 5963-N804)

Agent: NW

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1.0	∞	1.0	∞	1.0	∞	1.0	∞	1.0	∞	MΩ
C1-C18	Negative input clamp diode	III	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D18	Static output voltage, low level	IV	X	0.0	0.4	0 0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E18	Static output voltage, high level	V	X	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	V
F1-F18	Logic level conversion	VI	X	2 5	5.0	2 5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	V
G1-G18	Static output voltage, low level (V_{DD} = 15V)	VII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
H1-H18	Static output voltage, high level (V_{DD} = 15V)	VIII	X	13.5	15.15	13.5	15.15	13.5	15.15	13.5	15.15	13.5	15.15	V
J1-J18	Input current, high level	IX	X	0	200	0	200	0	200	--	--	0	200	nA
K1-K18	Input current, low level	X	X	-30	-45	-30	-45	-30	-45	--	--	-30	-45	µA
L1	Static power supply current, low level	XI	X	0	822	0	822	0	985	0	822	0	1170	µA
L2	Static power supply current, high level	XI	X	0	12	0	12	0	175	0	12	0	300	µA
M1-M18	Transition time, t_{TLH}, (C_L to 50 output)	XII	S	6	200	6	200	7	220	5	150	7	240	ns
M19-M36	Transition time, t_{TLH}, (C_L to 150 output)	XII	Q	12	400	12	400	13	440	9	305	15	475	ns
N1-N18	Transition time, t_{THL}, (C_L = 50pF)	XIII	S	2	80	2	80	2	90	2	60	3	97	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N19-N36	Transition time, t_{THL}, (C_L = 150pF)	XIII	Q	4	140	4	140	4	155	5	175	5	168	ns
P1-P18	Propagation delay time, t_{PLH}, (C_L = 50pF)	XIV	S	5	160	5	160	5	175	4	120	6	190	ns
P19-P36	Propagation delay time, t_{PLH}, (C_L = 150pF)	XIV	Q	6	210	6	210	7	232	5	160	9	247	ns
Q1-Q18	Propagation delay time, t_{PHL}, (C_L = 50pF)	XV	S	3	110	3	110	3	120	2	85	4	130	ns
Q19-Q36	Propagation delay time, t_{PHL}, (C_L = 150pF)	XV	Q	4	135	4	135	5	150	3	105	5	160	ns
R1-R18	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
S1-S18	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply transient, amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XIX	S	0.007	0.013	0 007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
V1-V18	Input capacitance	XX	Q	2	25	2	25	2	25	2	25	2	25	pF

NOTES:

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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Input voltage			
Low level			
V_{DD}= 5.0V -55°C	0 min, 1.5 max	0 min, 1.5 max	V
0° and 25°C	0 min, 1.5 max	0 min, 1 5 max	V
60°C	0 min, 1.5 max	0 min, 1.5 max	V
85°C	0 min, 1.5 max	0 min, 1.5 max	V
V_{DD}= 15.0V -55°C	0 min, 4.0 max	0 min, 4 0 max	V
0° and 25°C	0 min, 4.0 max	0 min, 4.0 max	V
60°C	0 min, 4.0 max	0 min, 4.0 max	V
85°C	0 min, 4.0 max	0 min, 4.0 max	V
High level			
V_{DD}= 15.0V -55°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
0° and 25°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
60°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
85°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
Output voltage (V_{DD} = 5.0V)			
Low level (I_{OL} = 3 75mA at -55°C)	0 4 min	0.4 min	V
(I_{OL} = 3.2mA at 0° and 25°C)	0.4 min	0.4 min	V
(I_{OL} = 2.68mA at 60°C)	0 4 min	0.4 min	V
(I_{OL} = 2.32mA at 85°C)	0 4 min	0.4 min	V
High level (I_{OH} = -1.85mA at -55°C)	2 5 min	2.5 min	V
(I_{OH} = -1.25mA at 0° and 25°C)	2 5 min	2.5 min	V
(I_{OH} = -1.08mA at 60°C)	2.5 min	2.5 min	V
(I_{OH} = -0.90mA at 85°C)	2.5 min	2.5 min	V
Output voltage (V_{DD} = 15V)			
Low level (I_{OL} = 30mA at -55°C)	1.5 min	1.5 min	V
(I_{OL} = 24mA at 0° and 25°C)	1.5 min	1.5 min	V
(I_{OL} = 20.6mA at 60°C)	1.5 min	1.5 min	V
(I_{OL} = 18.23mA at 85°C)	1.5 min	1.5 min	V
High level (I_{OH} = -5.5mA at -55°C)	13 5 min	13.5 min	V
(I_{OH} = -3.75mA at 0° and 25°C)	13 5 min	13.5 min	V
(I_{OH} = -3.26mA at 60°C)	13 5 min	13.5 min	V
(I_{OH} = -2.91mA at 85°C)	13.5 min	13.5 min	V

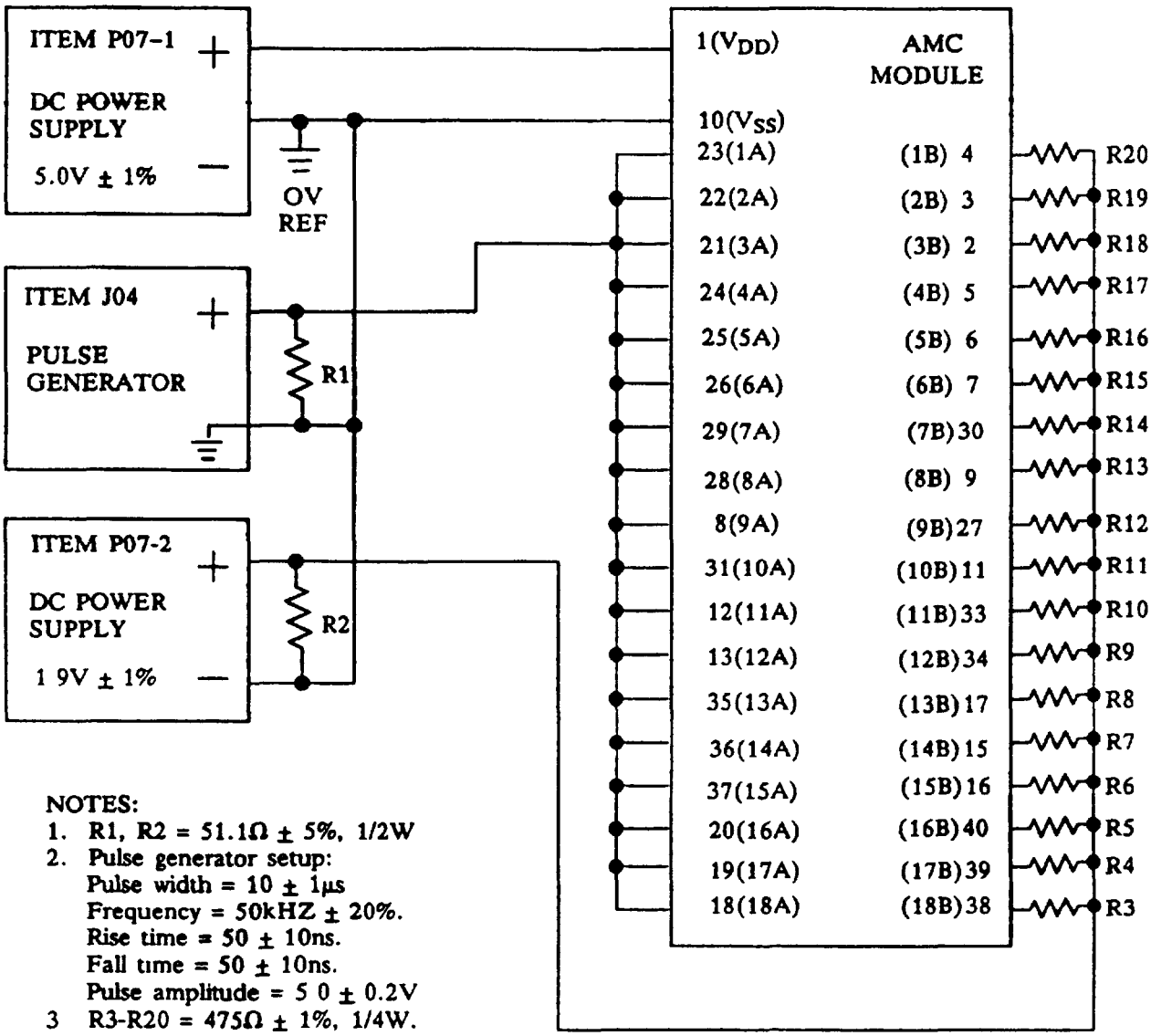
6.5 Module function. This contains 18 independent non-inverting buffers. The module may be used as a logic-level shifter if V_{DD} is the same as that of the driven TTL/DTL circuit. The input-signal high (V_{IH}) can exceed the V_{DD} supply voltage when these devices are used for logic level conversion.

6.5 1 Parallel outputs Paralleling of outputs is not permissible.

6.5.2 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge

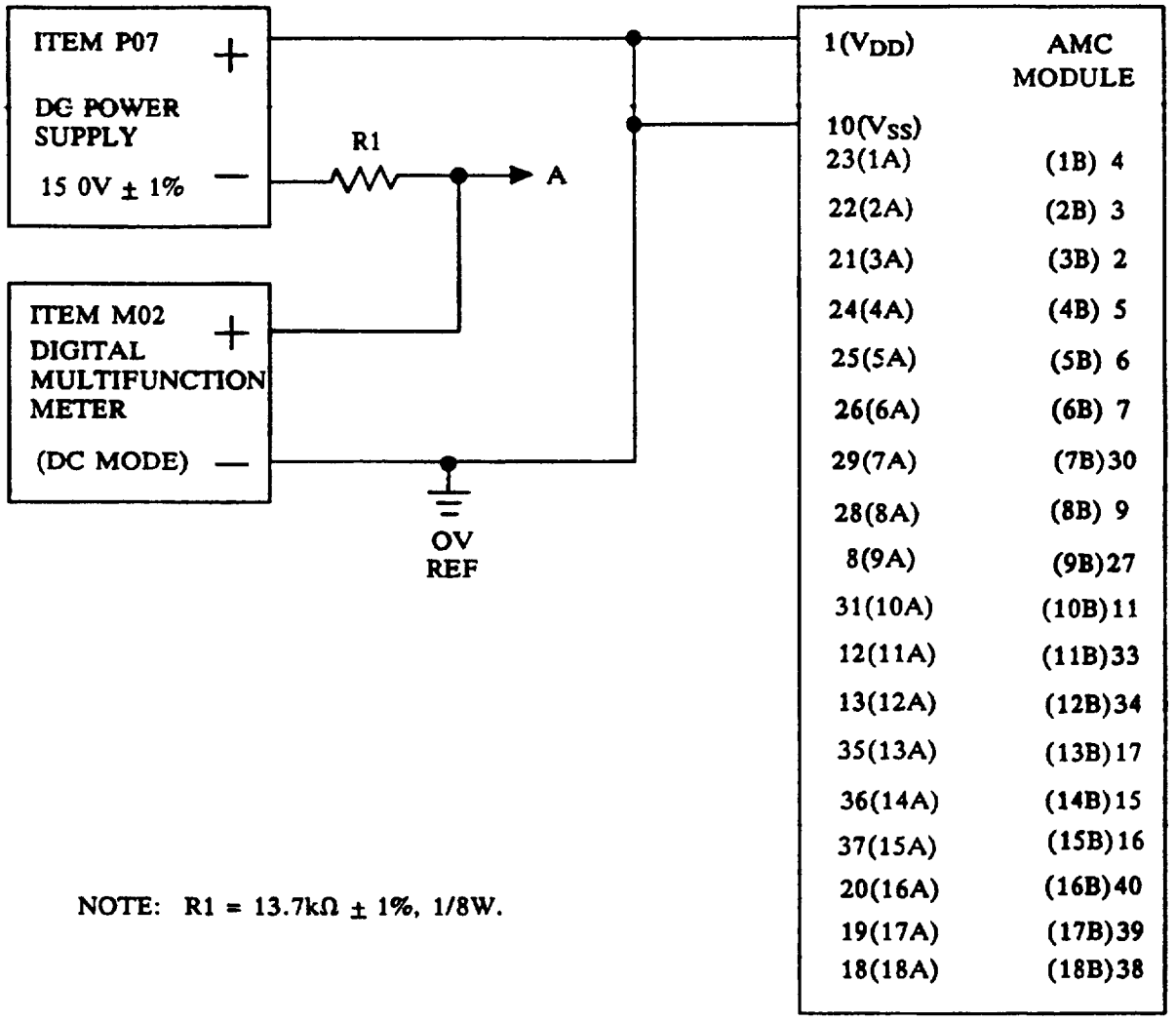
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NOTES:
1. R1, R2 = 51.1Ω ± 5%, 1/2W
2. Pulse generator setup:
   Pulse width = 10 ± 1μs
   Frequency = 50kHZ ± 20%.
   Rise time = 50 ± 10ns.
   Fall time = 50 ± 10ns.
   Pulse amplitude = 5 0 ± 0.2V
3. R3-R20 = 475Ω ± 1%, 1/4W.

FIGURE 3 Component temperature test circuit

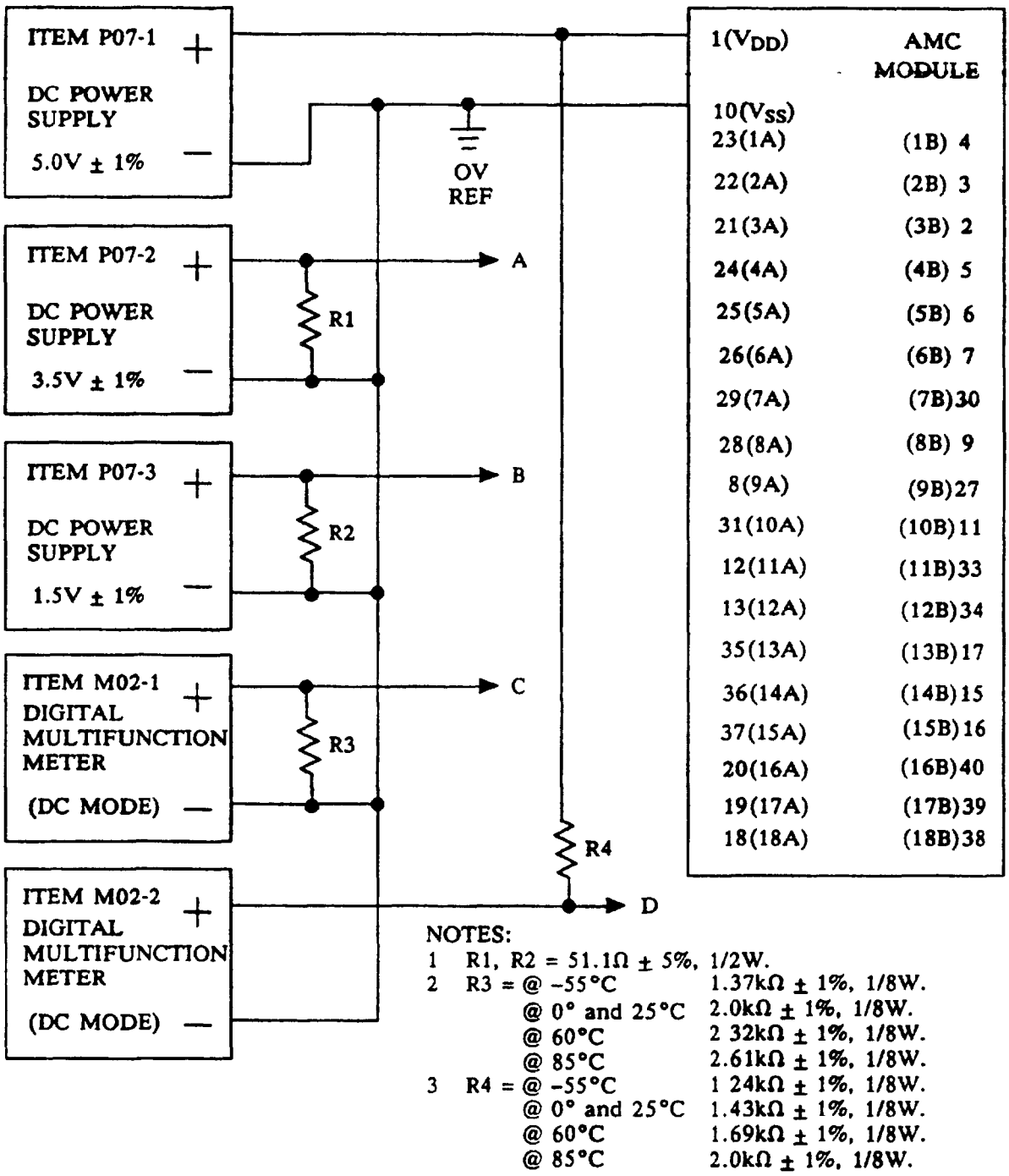
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NOTE: R1 = 13.7kΩ ± 1%, 1/8W.

FIGURE 6 Negative input clamp diode voltage test circuit for table III.

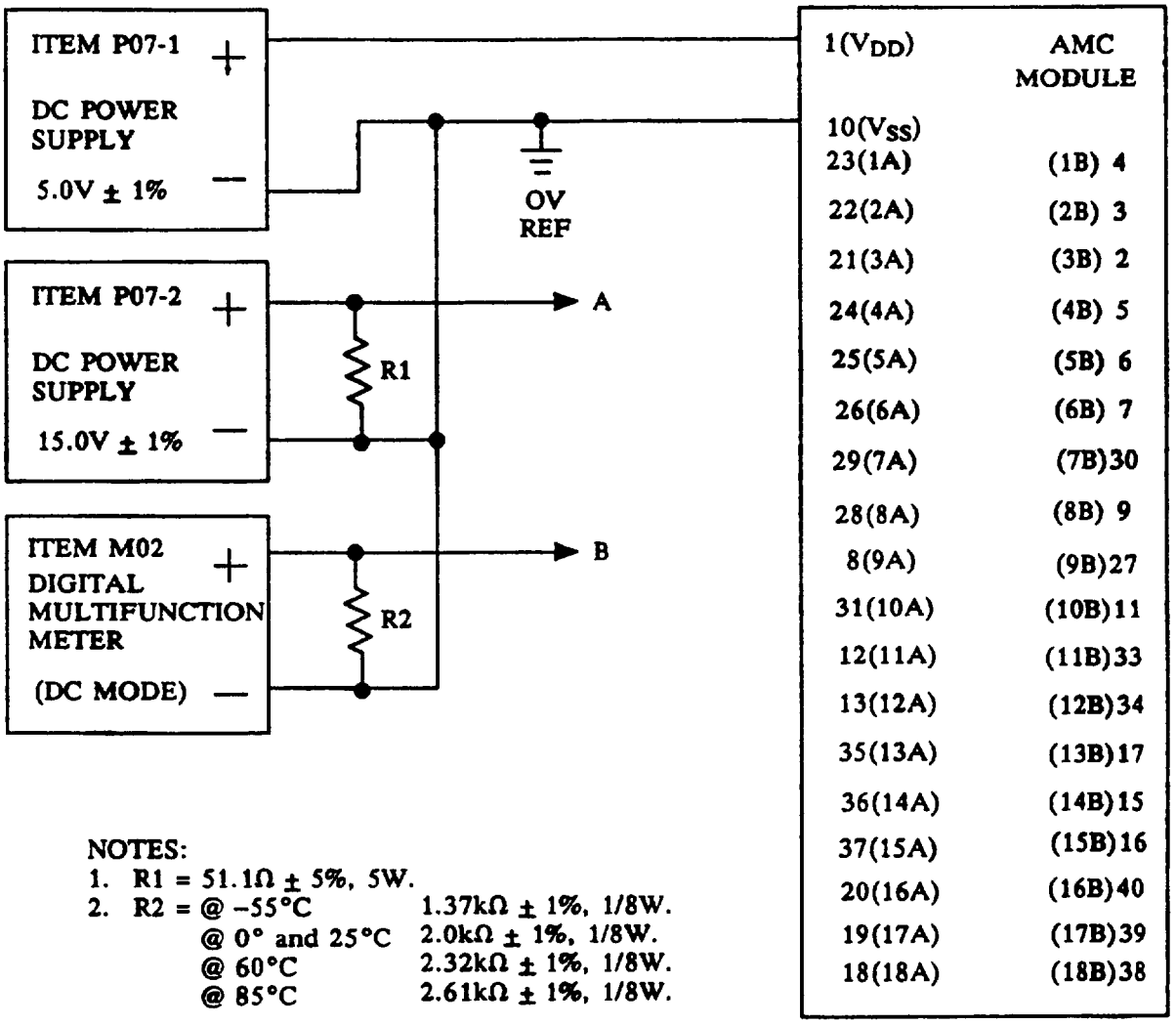
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NOTES:
1 R1, R2 = 51.1Ω ± 5%, 1/2W.
2 R3 = @ -55°C 1.37kΩ ± 1%, 1/8W.
@ 0° and 25°C 2.0kΩ ± 1%, 1/8W.
@ 60°C 2 32kΩ ± 1%, 1/8W.
@ 85°C 2.61kΩ ± 1%, 1/8W.
3 R4 = @ -55°C 1 24kΩ ± 1%, 1/8W.
@ 0° and 25°C 1.43kΩ ± 1%, 1/8W.
@ 60°C 1.69kΩ ± 1%, 1/8W.
@ 85°C 2.0kΩ ± 1%, 1/8W.

FIGURE 7. Static output voltage (low-level and high-level) test circuit for tables IV and V.

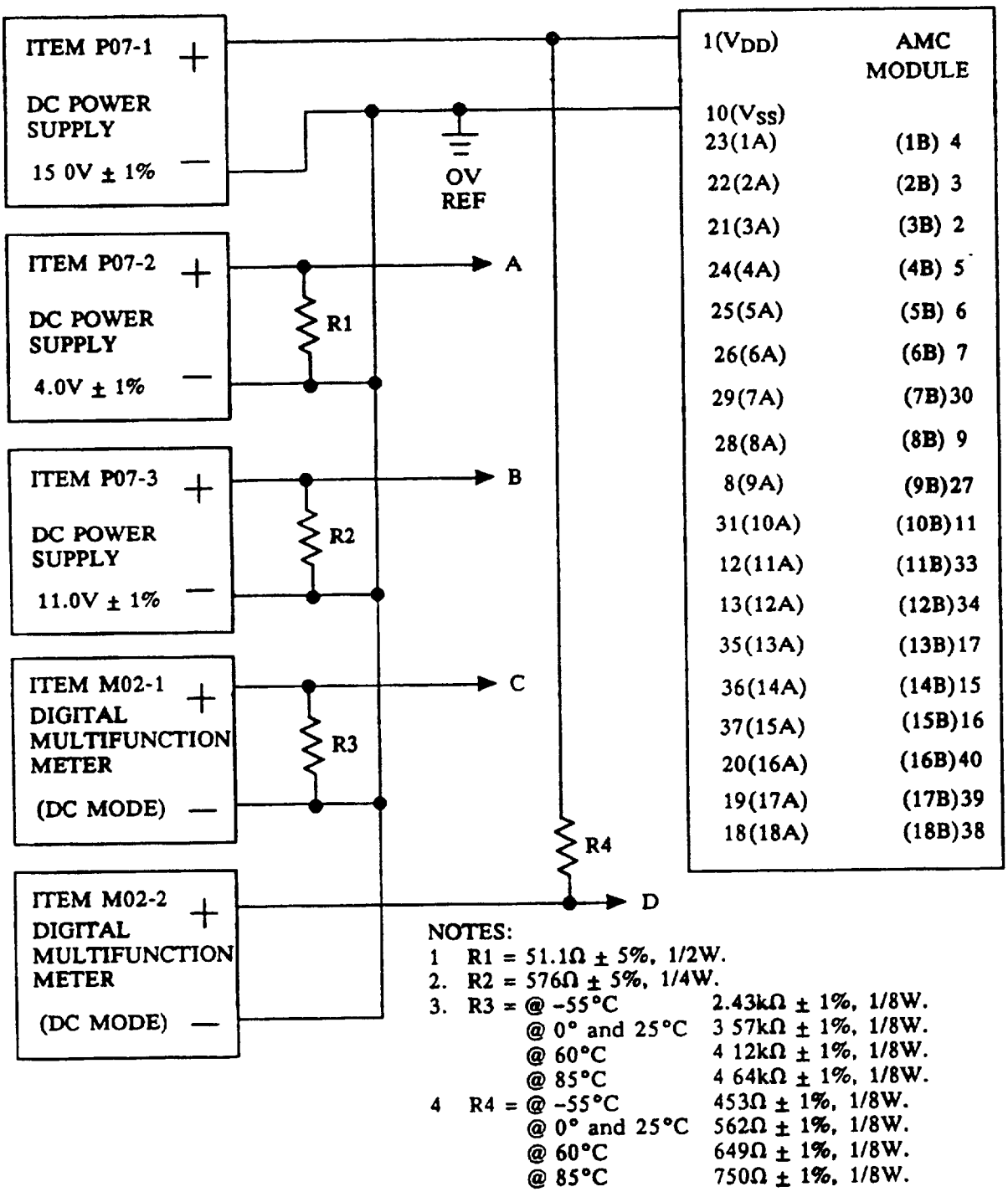
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NOTES:
1. R1 = 51.1Ω ± 5%, 5W.
2. R2 = @ −55°C 1.37kΩ ± 1%, 1/8W.
   @ 0° and 25°C 2.0kΩ ± 1%, 1/8W.
   @ 60°C 2.32kΩ ± 1%, 1/8W.
   @ 85°C 2.61kΩ ± 1%, 1/8W.

FIGURE 8. Logic level conversion test circuit for table VI

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NOTES:
1. R1 = 51.1Ω ± 5%, 1/2W.
2. R2 = 576Ω ± 5%, 1/4W.
3. R3 = @ -55°C 2.43kΩ ± 1%, 1/8W.
   @ 0° and 25°C 3 57kΩ ± 1%, 1/8W.
   @ 60°C 4 12kΩ ± 1%, 1/8W.
   @ 85°C 4 64kΩ ± 1%, 1/8W.
4. R4 = @ -55°C 453Ω ± 1%, 1/8W.
   @ 0° and 25°C 562Ω ± 1%, 1/8W.
   @ 60°C 649Ω ± 1%, 1/8W.
   @ 85°C 750Ω ± 1%, 1/8W.

FIGURE 9 Static output voltage (low-level and high-level) test circuit for tables VII and VIII.

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